

Advanced (Habanero)

- Q1.** Which of the following is the point-slope form of a linear equation?
- (A) $y - y_1 = m(x - x_1)$
 (B) $y = mx + b$
 (C) $Ax + By = C$
 (D) $x = my + b$
 (E) $y = \frac{1}{m}x + b$
- Q2.** What is the slope-intercept form of the equation $2x + 3y = 7$?
- (A) $y = -\frac{2}{3}x + \frac{7}{3}$
 (B) $y = \frac{2}{3}x + \frac{7}{3}$
 (C) $y = \frac{3}{2}x + \frac{7}{2}$
 (D) $y = -\frac{3}{2}x + \frac{7}{2}$
 (E) $y = \frac{7}{2}x - \frac{3}{2}$
- Q3.** Which equation represents a line with a slope of -2 and a y -intercept of 4 ?
- (A) $y = -2x + 4$
 (B) $y = 2x - 4$
 (C) $y = -2x - 4$
 (D) $y = 2x + 4$
 (E) $x = -2y + 4$
- Q4.** What is the standard form of the equation $y = \frac{3}{4}x - 2$?
- (A) $3x - 4y = 8$
 (B) $4x - 3y = 8$
 (C) $3x + 4y = 8$
 (D) $4x + 3y = 8$
 (E) $-3x + 4y = 8$
- Q5.** The equation $y = 2x - 3$ represents a line that passes through the point $(2, 1)$. What is the slope of this line?
- (A) $m = 2$
 (B) $m = -2$
 (C) $m = 1$
 (D) $m = -1$
 (E) $m = 0$
- Q6.** The equation $x + 2y = 5$ can be written in slope-intercept form as
- (A) $y = -\frac{1}{2}x + \frac{5}{2}$
 (B) $y = \frac{1}{2}x + \frac{5}{2}$
 (C) $y = -\frac{2}{1}x + \frac{5}{1}$
 (D) $y = \frac{2}{1}x + \frac{5}{1}$
 (E) $y = \frac{5}{2}x - \frac{1}{2}$
- Q7.** A line has a slope of 3 and passes through the point $(1, 2)$. What is the equation of this line in point-slope form?
- (A) $y - 2 = 3(x - 1)$
 (B) $y - 2 = -3(x + 1)$
 (C) $y + 2 = 3(x - 1)$
 (D) $y - 1 = 3(x - 2)$
 (E) $y + 1 = 3(x + 2)$
- Q8.** The equation $y = \frac{2}{3}x - 1$ represents a line with a y -intercept of
- (A) $(-1, 0)$
 (B) $(-1, 1)$
 (C) $(0, -1)$
 (D) $(1, 0)$
 (E) $(0, 1)$

Open Ended Questions (FRQ)

- Q9.** Find the equation of the line that passes through the points $(2, 3)$ and $(4, 5)$. Write your answer in slope-intercept form.
- Q10.** A water tank can hold 1200 gallons of water. Due to a leak, the tank is losing water at a rate of 2 gallons per minute. If t represents the time in minutes and V represents the volume of water in gallons, write an equation that represents the volume of water in the tank after t minutes. Graph this equation and find the time at which the tank will be empty.

Chef's Tips

- Make sure to check the units of the variables.
- Graphing calculators are not allowed.
- Show all work and explain your reasoning.